

Experimental and numerical simulation of water movement in soil

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Preferential flow refers to a flow transport pattern where water and solute flows around the soil matrix, contributing to accelerating velocity of movement in the soil; soil recharge occurs over the whole inlet boundary. Preferential flow is produced by a large number of macropores and fractures in the soil, such as root or worm holes, and is very important to groundwater recharge owing to its rapid movement. Using the reaction of iodine turning blue on contact with starch, outdoor tracer experiments of heterogeneous flow in clay loam for different scales and boundary conditions are designed, so that the heterogeneous flow patterns can be studied. Using experimental image analysis, the soil is divided into a matrix and a preferential channel network, formed by the spatial structure of the flow channels. An irregular flow network is established through random generation of angles, lengths, aperture and location coordinates of flow channels, in order to achieve an accurate description of preferential flow channels. According to the soil moisture of each layer obtained by tracer experiments, soil water retention curve and saturated hydraulic conductivity, soil moisture variation was simulated. The distribution of the channel network and models of water movement in the soil are used to undertake a quantitative analysis of the influence of preferential flow on soil water migration.

Notation

A	volume of influence for fracture node I [L^3]	q^*	fluid exchange with the outside of the simulation domain, as specified from boundary conditions
$(2b)$	fracture aperture [L]	S_w	subsurface water saturation [dimensionless]
dB	boundary of volume V in boundary B	S_{wf}	fracture water saturation [dimensionless]
g	gravitational acceleration [LT^{-2}]	t	time [T]
K_f	saturated soil water conductivity of channel flow [LT^{-1}]	V	region or control volume associated with a node [L^3]
K_{ij}	anisotropic components of saturated soil water conductivity	x_i ($i = 1, 2, 3$)	spatial coordinates where $x_1 = x$ is the x -coordinate, $x_3 = z$ is the vertical coordinate
K_{rw}	relative permeability of the medium with respect to the degree of water saturation	z	subsurface elevation head [L]
N_I	standard finite element basis functions	z_f	fracture elevation head [L]
$P(\alpha) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	normal distribution $P(x)$ for a variable x with mean μ and variance σ^2	Γ_{IJ}	term describing fluid flow between nodes I and J
$P(x) = P_0e^{-\lambda x}$	exponential distribution for a variable x with rate parameter λ	γ	pressure head difference between nodes I and J [L]
$P(x) = \frac{1}{\sigma x\sqrt{2\pi}}e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$	log-normal distribution for a variable x with mean μ and variance σ^2	Δt	time step [T]
Q	exchange of water between two domains [T^{-1}]	θ_r	residual moisture content [$L^3 L^{-3}$]
Q_f	exchange of water between the flow channel and matrix domains [T^{-1}]	θ_s	saturated soil water content [$L^3 L^{-3}$]
		λ	upstream weighting
		ρ	water, density [ML^{-3}]
		ψ	pressure head [L]

Sub- and superscripts

f	fracture
I, J	denote a property of node I or J respectively
i, j	spatial indices
L	denotes a time level in a time-stepping procedure

1. Introduction

The unsaturated zone plays a critical role for various components of the hydrological cycle, such as infiltration, soil moisture storage, evaporation, plant water uptake, groundwater recharge, runoff and erosion. As a result, groundwater flow and solute transport in the unsaturated zone have been investigated in soil science for decades. It has long been demonstrated that water flow in soils can either be uniform (Green and Ampt, 1911) or non-uniform (Lawes *et al.*, 1881). Uniform flow leads to stable wetting fronts that are parallel to the soil surface, while non-uniform flow results in irregular wetting of a soil profile as a direct consequence of water moving faster in certain parts of the soil profile than in others. This non-uniform movement of water and dissolved solutes is commonly denoted as preferential flow and the study of preferential flow and transport has received great attention in soil and agricultural sciences because of its implications in accelerating the movement of agricultural contaminants (fertilisers, pesticides, pathogenic microorganisms, toxic trace elements) through the unsaturated zone and then to the underlying groundwater (Hendrickx and Flury, 2001). Hendrickx and Flury (2001) defined preferential flow as ‘all phenomena where water and solutes move along certain pathways, while bypassing a fraction of the porous matrix’. Thus, an important characteristic of preferential flow is that during wetting, part of the moisture front can propagate quickly to significant depths while bypassing a large part of the matrix pore space. Water and solutes may move to far greater depths, and much faster, than would be predicted with Richards’ equation using area-averaged moisture contents and pressure heads (Beven, 1991).

A number of studies of dye-stained preferential pathways have been published in the literature (e.g. Forrer *et al.*, 2000; Morris and Mooney, 2004; Wang *et al.*, 2006, 2009, 2010; Yasuda *et al.*, 2001). However, because of dye adsorption onto the soil matrix and its large molecule size, the dye-stained soil front may lag behind the actual wetting front (Glass *et al.*, 2005; Nobles *et al.*, 2010). From distributions of combined dye (Brilliant Blue FCF) and bromide tracers in a layered soil, Kasteel *et al.* (2002) pointed out that the dye and bromide did not follow the same flow pathways. Therefore, dye may be unsuitable to trace the soil water paths, especially in soils with high clay content. Wang *et al.* (2006, 2009, 2010) characterised the macropore structure as well as heterogeneous soil water flow patterns using combined tracers of dye and iodine. In a clayey soil, dye can penetrate into deeper soil only along the macropores and the dye-stained patterns were therefore used to describe the macropore structure,

while the iodine-stained patterns were used to visualise the heterogeneous flow patterns.

Several models that simulate preferential flow have been developed in the past two decades. Those models are generally based on either the dual-permeability, kinematic wave, or two-stage conceptual approaches, which all assume mobile and immobile regions in a soil. The models differ in their description of flow and transport in the preferential flow domain and of mass transfer. For example dual-porosity and dual-permeability models both assume that the porous medium consists of two interacting regions, one associated with the inter-aggregate, macropore, or fracture system, and one comprising micropores (or intra-aggregate pores) inside soil aggregates or the rock matrix. Dual-porosity models assume that water in the matrix is stagnant, but dual-permeability models assume that water can move in both the inter- and intra-aggregate pore regions (Gerke and van Genuchten, 1993; Jarvis, 1994; Pruess and Wang, 1987). Available dual-permeability models differ mainly in how they implement water flow in and between the two pore regions, especially with respect to the degree of simplification and empiricism (Simunek *et al.*, 2003). For the application of these models, it is necessary to know the properties of soil. So the uses of these models are obviously confined by irregular distribution of macropores and fractures in the soil.

This paper presents results from field injection of water containing a tracer into a soil. The migration of the tracer was observed with image analysis, which provided flow pathways at a resolution higher than previously achieved. The concept of the water flow channel was introduced to represent the spatial distribution of water flow pathways in the soil. The irregular flow network is established through generation of fractures having random angles, lengths, aperture, and location. To accurately describe preferential flow, the soil is divided into a matrix and a preferential flow network, which is formed by the spatial structure of the fracture network. The equations describing flow in the soil matrix and in the irregular water network are coupled to account for the flow mutual influence caused by the cross-fracture network, and also to improve computational efficiency.

2. Experiment and methods

2.1 Heterogeneous flow experiment

The iodide ion (I^-) has transport properties similar to those of the bromide (Br^-) ion because of their chemical similarity (Bowman, 1984). When mixed with starch, the I^- ion can be easily visualised, which is the method used for outdoor tracer experiments of heterogeneous flow in clay loam of different scales and different boundary conditions designed at the Irrigation, Drainage and Water Environment station, Wuhan University in 2007.

For the tracer experiments, Fe^{3+} was used as a mild oxidiser for I^- , which will not bleach the formed starch-iodide complex

colour. In addition, starch was dissolved in the I^- solution rather than applying starch powder directly to the soil profile to avoid the wetting process of the starch layer by I^- solution in soil. After these modifications, the concentration of the iodide tracer solution was considerably lowered, from an original value of 5–15 g/l down to 2–3 g/l. Its indication solution consists of 50 g/l of starch and 0.05M iron (III) nitrate ($Fe(NO_3)_3$). After spraying, it took about 10 min for a dark blue-violet colour to develop as a result of the oxidiser iron (III) nitrate.

To prepare the 1.0 litre iodide indication solution, 50 g of water-soluble linear starch is first wetted with 0.05 litre of distilled water, and then mixed with 0.8 litre of boiling distilled water. The resulting solution is stirred until it becomes clear. The solution is then cooled to room temperature and 20.20 g of $Fe(NO_3)_3 \cdot 9H_2O$ is added and stirred until it dissolves, to bring the total volume to 1.0 litre. The resulting solution is stored in a brown glass bottle and remains stable for at least 30 days.

To trace water movement in soils, water containing the iodide tracer is applied on the soil in the area of concern. After infiltration and redistribution of the tracer cease, a vertical soil profile is excavated and smoothed with a flat shovel. The indication solution is then added with a pressurised sprayer that can release a continuous and fine mist of indication solution. The indication solution is sprayed onto the surface as uniformly as possible by long, slow strokes. The colour changes occur gradually after spraying.

In total five experiments were conducted under different conditions. The experimental set-up consisted of four plots covering an area of $1.0 \times 1.0 \text{ m}^2$ each, which were irrigated with 20 mm, 40 mm, 60 mm and 80 mm depths of potassium iodide solution respectively, and a $2.0 \times 1.0 \text{ m}^2$ plot irrigated with 40 mm depths of potassium iodide solution. Profiles were excavated every 5 cm in the x direction (Figure 1). In each profile, an oxidiser liquor was sprayed to form the iodide–starch blue complex, then vertical core samples were taken by digital camera. Bulk samples

were also taken and analysed for electric conductivity and water content by vertically removing surfacial soil.

The resulting pictures of the tracer experiments (Figure 2) illustrate that the heterogeneous flow pattern in the soil can be represented by a network of channels, or fractures. As a result of the connectivity of this channel network, water and solutes move along given pathways and bypass parts of the porous matrix. Another important feature is the water exchange between matrix area and flow networks, when water moves along the flow channels at the same time.

2.2 Image processing

The most important sources of variability in the analysis of soil images are uneven illumination, geometric distortion and the condition of the surface (Aeby *et al.*, 1997; Forrer *et al.*, 2000). These problems were minimised here by fixed environmental conditions and carefully preparing by hand the soil face prior to the acquisition of each image. Grey scales were positioned at the sides of the image for corrections of white balances. The exposed soil faces were prepared carefully by hand by removing thin layers of the soil with a knife. High-resolution (1024×1024 pixels) digital images were collected in the BMP file format. Corrections were made for white balance between images in order to standardise the images for comparison.

The threshold algorithm (Baveye *et al.*, 1998) was used to transform colour images into black and white (binary) images to classify images with reference to stained and unstained areas of the soil (see Figure 3). In other words, any pixels with red, green and blue values that fall within the specified range were detected.

The flow patterns for five groups are shown in Figure 4. Grids having an area of $10 \text{ cm} \times 10 \text{ cm}$ were placed onto the soil profile images. The flow pattern varies depending on the injection conditions. The flow patterns consisted mainly of small-area preferential flow for the 20 mm irrigated solution, while rapid and bypassing preferential flow was the main transport mechanism when the amounts of water increased from 40 mm to 60 mm. The flow patterns were part matrix flow and overall matrix flow when the water infiltration reached 80 mm.

3. Model description

3.1 Introduction to the model

The hydrogeosphere model (Graf and Therrien, 2007; Therrien *et al.*, 2004) is an efficient and robust numerical model that solves the three-dimensional variably saturated subsurface flow and solute transport equations in non-fractured or discretely fractured media. It can simulate water flow within discrete fracture as well as water flow in the soil matrix based on the dual-permeability concept using the equivalent freshwater formulation.

The porous low-permeability matrix is represented by regular three-dimensional (3D) blocks and fractures of high permeabil-

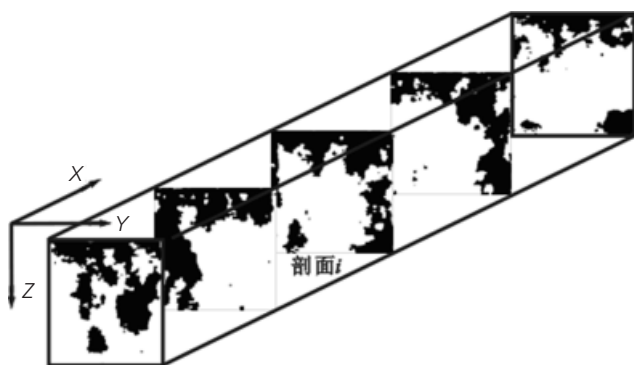


Figure 1. Diagram of the experimental set-up

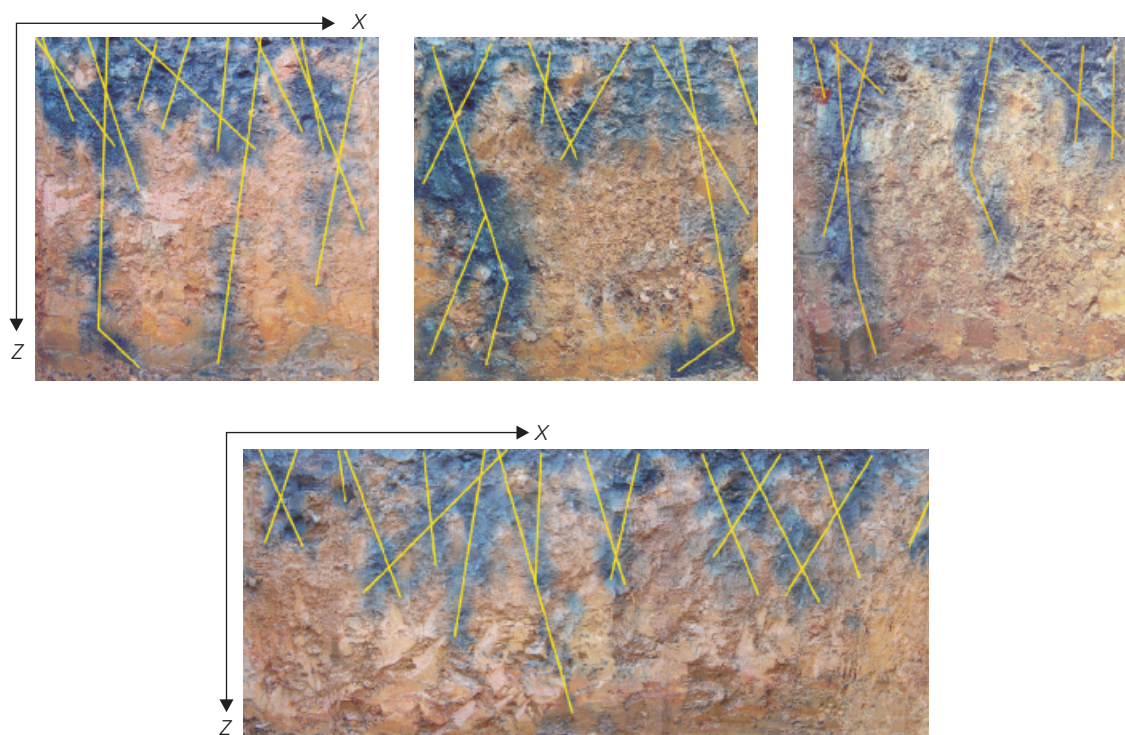


Figure 2. Experimental analyses of flow channels

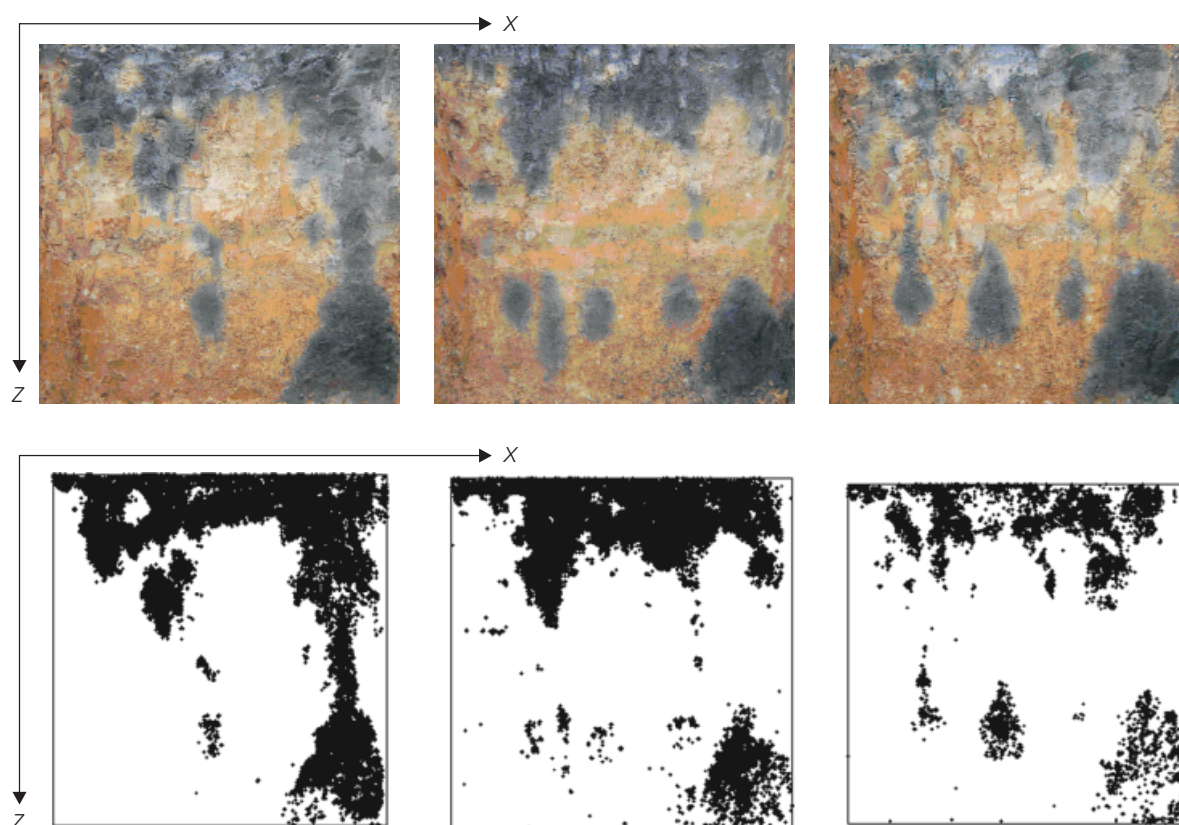


Figure 3. Heterogeneous dye coverage patterns at experiment plot before and after image processing

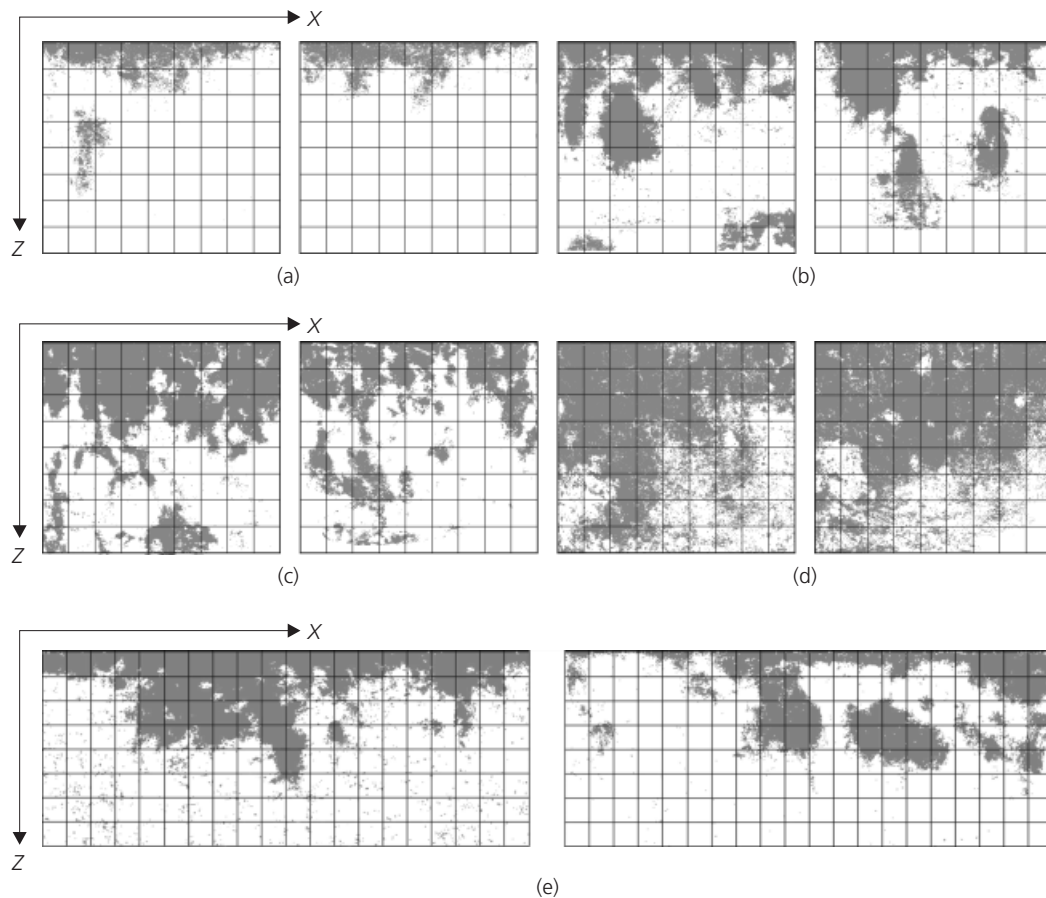


Figure 4. Flow patterns under various conditions:

- (a) experiment 1, $x = 30$ cm (left) and $x = 70$ cm (right);
- (b) experiment 2, $x = 30$ cm (left) and $x = 70$ cm (right);
- (c) experiment 3, $x = 30$ cm (left) and $x = 70$ cm (right);
- (d) experiment 4, $x = 30$ cm (left) and $x = 70$ cm (right);
- (e) experiment 5, $x = 30$ cm (left) and $x = 70$ cm (right)

ity are represented by two-dimensional (2D) rectangular planes. Fluid flow and transport equations are solved for both porous matrix and fractures. Vertical, horizontal and inclined 2D fractures are incorporated into the 3D grid by superimposing 2D fracture elements onto 3D regular block elements. Faces and blocks share common nodes along the fracture walls. Thus, nodes at fracture locations receive fluid and mass contributions from both the block elements, as well as from the fracture faces. Furthermore, for these mutual nodes, both hydraulic head and concentration at the fracture–matrix interface are assumed to be equal, ensuring continuity of these variables between fractures and matrix (Graf and Therrien, 2007; Therrien *et al.*, 2004).

In the following, a detailed description is provided of the model theory for the processes of initiation and interaction and how they are combined within the hydrogeosphere model.

3.2 Construction of flow networks

There are different ways to computationally construct a flow network. In this study, simple 2D flow networks were constructed by generating fractures with random size, orientation, location and conductivity. The procedure used to construct the flow network includes several steps.

First, as the fractures in soils mainly result from shrinkage and expansion, the growth of plant roots and burrows formed by earthworm activity in soils, these fractures should be downward from the soil surface. A small number of points were randomly generated along a line representing the ground surface. Orientations, channel lengths and aperture distribution were considered to be independent during the numerical procedure for generating flow networks.

Second, for each randomly generated point, fractures (with the

randomly generated point as the initial point) were generated. The lengths of the fractures were randomly determined using a log-normal distribution according to

$$1. \quad P(x) = \frac{1}{\sigma x \sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

The default fracture orientation follows a double-peaked Gaussian distribution, according to

$$2. \quad P(\alpha) = P_1(\alpha) + P_2(\alpha)$$

where the normal distribution $P(x)$ for a variable x with mean μ and variance σ^2 has the probability function

$$3. \quad P(\alpha) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad x \in (-\infty, \infty)$$

Third, the discharge capacity of fracture is determined by its hydraulic conductivity K , which depends on its aperture. It is therefore necessary to simulate the aperture distribution of fractures.

The channel aperture distribution follows an exponential distribution, according to

$$4. \quad P(x) = P_0 e^{-\lambda x}$$

The aperture of each natural channel varies in a complex manner. For example, the radius of macropores in soils gradually decreases with depth. The simulation of the aperture of a channel must also take into account the general law, that the aperture decreased from top to bottom, the rate of reduction was randomly generated (Graf and Therrien, 2007; Therrien *et al.*, 2004).

3.3 Governing equations

The following modified form of Richards' equation is used to describe 3D transient subsurface flow in a variably saturated porous medium

$$5. \quad \frac{\partial}{\partial x_i} \left[K_{ij} K_{rw} \frac{\partial(\psi + z)}{\partial x_j} \right] \pm Q = \frac{\partial}{\partial t} (\theta_s S_w)$$

where θ_s ($\text{l}^3 \text{l}^{-3}$) is the saturated soil water content, S_w is the saturation, θ_r ($\text{l}^3 \text{l}^{-3}$) is the residual moisture content, ψ (litre) is the pressure head, Q (T^{-1}) is the exchange of water between two domains. x_i ($i = 1, 2, 3$) are the spatial coordinates where $x_1 = x$ is the x coordinate, $x_3 = z$ is the vertical coordinate, t (T) is time, K_{ij} are the anisotropic components of saturated soil water

conductivity, K_{rw} represents the relative permeability of the medium with respect to the degree of water saturation S_w .

A flow channel is idealised here as the space between 2D parallel surfaces, with the tacit assumptions that the total head is uniform across the channel width. The cubic law can be used to describe channel flow as follows

$$6. \quad \begin{aligned} \frac{\partial}{\partial x_i} (2b) K_f k_{rwf} \frac{\partial(\psi_f + z_f)}{\partial x_j} &\pm Q_f \\ &= (2b) \frac{\partial S_{wf}}{\partial t} \end{aligned}$$

where Q_f (T^{-1}) is the exchange of water between the flow channel and matrix domains, K_f is saturated soil water conductivity of channel flow, K_{rw} represents the relative permeability of the medium with respect to the degree of water saturation S_w .

$$7. \quad K_f = \frac{\rho g (2b)^2}{12\mu}$$

3.4 The coupling of two-domain equations

A control volume finite-element method is used to spatially discretise the flow equations. The discrete flow equation for the matrix domain is

$$8. \quad \begin{aligned} \left[\theta_s \frac{(s_w^{L+1} - s_w^L)}{\Delta t} \right]_I V_I &= \sum_{J \in \eta_I} (\gamma_{IJ} \lambda_{IJ})^{L+1} \Gamma_{IJ} \\ &+ Q_I^{L+1} V_I + \int_{B^e} q^* N_I dB^e \end{aligned}$$

Faces and blocks share common nodes along the flow channel walls. Thus, nodes at fracture locations receive fluid and mass contributions from both the block elements and from the fracture faces. Furthermore, for these mutual nodes, both hydraulic heads at the fracture–matrix interface are assumed to be equal, ensuring continuity of these variables between fractures and matrix (Graf and Therrien, 2007; Therrien *et al.*, 2004).

In Equation 8, $Q_I^{L+1} V_I$ is the value of the fluid exchange between the domains, and in the discrete flow channel equation

$$9. \quad \begin{aligned} (2b) \frac{(\hat{S}_{wf}^{L+1} - \hat{S}_{wf}^L)}{\Delta t} A_I \\ = (2b) \sum_{J \in \eta_{fI}} (\gamma_{fIJ} \lambda_{fIJ})^{L+1} \Gamma_{IJ} + Q_I^{L+1} A_I \end{aligned}$$

As the hydraulic conditions in channel networks and matrix domains are quite different, it is inevitable that interaction of water and solution transport in matrix and macropore will take place. Water exchange is the crucial factor for coupling the two domains. And $Q_I^{L+1} A_I$ is exactly the value of the fluid exchange between the domains, so Equation 9 can be rewritten as

$$10. \quad Q_I^{L+1} A_I = (2b) \frac{(\hat{S}_{wf}^{L+1} - \hat{S}_{wf}^L)}{\Delta t} A_I - (2b) \sum_{J \in \eta_{fI}} (\gamma_{fIJ} \lambda_{fIJ})^{L+1} \Gamma_{IJ}$$

Replacing the exact value of the fluid exchange into Equation 10 gives the following two-domain coupling equation

$$11. \quad \begin{aligned} & [(\theta_s S_w)_I^{L+1} - (\theta_s S_w)_I^L] \frac{V_I}{\Delta t} \\ & + (2b) [(S_{wf})_I^{L+1} - (S_{wf})_I^L] \frac{A_I}{\Delta t} \\ & = \sum_{J \in \eta_{fI}} [(\gamma_{fIJ} \lambda_{fIJ})^{L+1} \Gamma_{IJ} + Q_I^{L+1}] \\ & + \left[(2b) \sum_{J \in \eta_{fI}} (\gamma_{fIJ} \lambda_{fIJ})^{L+1} \Gamma_{fIJ} + Q_I^{L+1} \right] \\ & + \int_{B^e} q^* N_I dB^e \end{aligned}$$

4. Boundary conditions and parameters estimation

4.1 Initial and boundary conditions

In order to verify the accuracy of the hydrogeosphere model, a series of field experiments was simulated for a uniform x -direction, 2D flow field. In addition to domain-internal heterogeneity, the geometry of field boundaries may be irregular, the soil surface shaped, and the boundary conditions spatially distributed. While field-scale transport processes are essentially 3D, they may, at times, be simplified and reduced to 2D descriptions.

The simulation domains were rectangles with dimensions of $1.0 \text{ m} \times 1.0 \text{ m}$ and $2.0 \text{ m} \times 1.0 \text{ m}$ respectively. The left and right boundaries were assumed to be no-flow boundaries, the top boundary was assumed to have a water head of 60 mm and 40 mm corresponding to the $1.0 \text{ m} \times 1.0 \text{ m}$ and $2.0 \text{ m} \times 1.0 \text{ m}$ domains, respectively, and the bottom boundary condition was a free drainage boundary. Experimental data of soil water content were used for the calculation initial conditions.

4.2 Hydrodynamics and fracture parameters

The saturated hydraulic conductivity is the crucial parameter to simulate fluid flow in unsaturated soil; meanwhile, it can forecast unsaturated soil hydraulic conductivity. The experimental soil has clay properties, and the soil hydraulic parameters are listed in Table 1; the hydraulic parameters of fracture can be calculated by Equation 7.

The default parameters for the irregular channel network, such as angles, lengths, widths, location of flow channels, were extracted from the $1.0 \text{ m} \times 1.0 \text{ m}$ experiment. To evaluate the influence of the network on the result, 100 groups of irregular networks were calculated. Figure 5 shows the fitting results and Figure 6 shows two different groups of flow channels with the same parameters under 1.0 m wide. The parameters for the flow channels are listed in Table 2. In the domain, which was 1.0 m wide, the total number of flow channels was set as 40. As the scale of width is enlarged to 2.0 m , the parameters for the flow channels were unchanged while the number of channels was raised to 80.

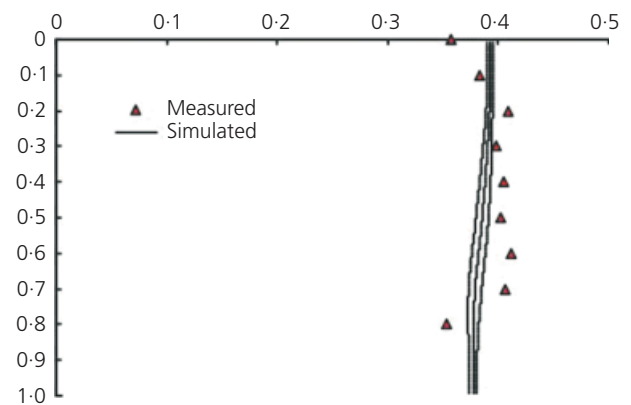


Figure 5. Comparison of experimental data (Δ) and simulation data of water content under 1.0 m width

Soil particle size distribution: %			Bulk density: g/cm^3	Porosity: %	K : m/day	θ_r	θ_s	β	α : per m
>50 μm	2–50 μm	<2 μm							
4.225	52.15	44.45	1.508	39.1	0.098	0.068	0.4	1.09	0.8

Table 1. Soil physical and hydraulic properties

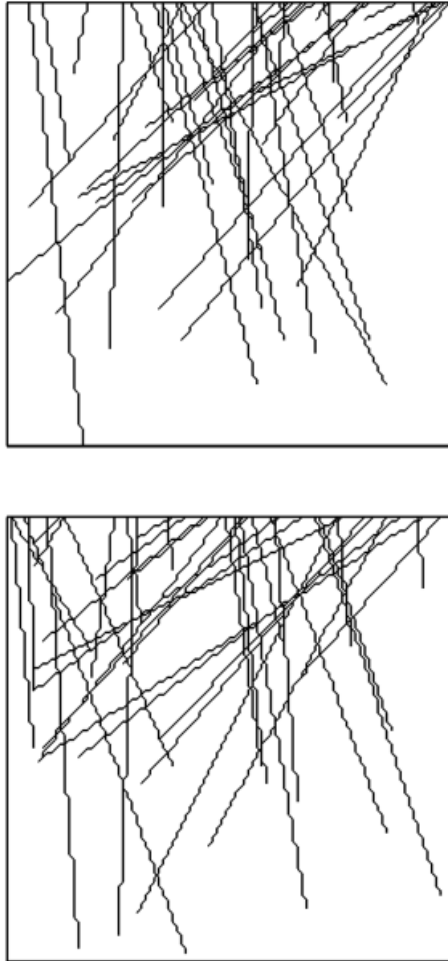


Figure 6. Two different groups of flow channels with the same parameter under 1.0 m width

5. Results and discussion

5.1 Simulation results

In the numerical model, medium properties and flow boundary conditions were set up as described above. The 2.0 m wide experimental and simulation data of soil water content are shown in Figure 7. To analyse the results of different networks, different, yet statistically equivalent networks have also been investigated. In total, 100 seeds were used to generate 100 groups of random structured fracture networks.

For comparison, appropriate field data are necessary to validate the model. The calculation results show that the variance of calculation for 100 groups converged. Simulated results of the soil profile moisture compared well with experimental results. However, in the experiment the same depths of different profiles were measured at different times. Furthermore, evaporation was

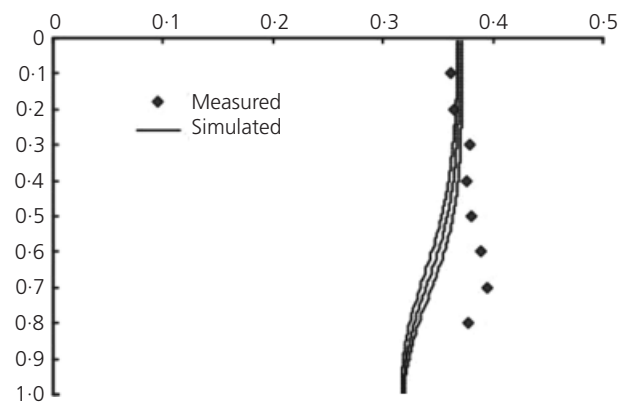


Figure 7. Comparison of experimental data (Δ) and simulation data of water content under 2.0 m width

Values for fracture orientation			Values for fracture length			Values for fracture aperture		
Parameter	Value	Unit	Parameter	Value	Unit	Parameter	Value	Unit
Number of orientation classes: degrees	13		Number of length classes	10		Number of aperture classes	10	
Middle of the smallest orientation class	30	degrees	Middle of the smallest length class	0.1	m	Middle of the smallest aperture class	50	mm
Middle of the largest orientation class	150	degrees	Middle of the largest length class	1.0	m	Middle of the largest aperture class	300	mm
Standard deviation of both classes	15	degrees	μ of the log-normal distribution	2.9		λ of the exponential aperture distribution	9000	
Mean of the first Gaussian distribution μ_1	60	degrees	σ of the log-normal length distribution	0.45				
Mean of the second Gaussian distribution μ_2	120	degrees						

Table 2. Default values for 2D random fracture

not considered during the experiments, which explains why simulated results are slightly higher than experimental results.

Figure 8 summarises the generated channel networks, which differ from each other as a result of various seeds used, although their generation parameters are the same. However, the differences in the channel network have little influence on calculation results after comparing experimental and simulation data.

The distribution of the channel network can be seen in Figure 9, while Figure 10 shows the dynamic process of soil water movement based on the channel networks shown in Figure 9. The colour bar on the right-hand side indicates soil water content. The images illustrate that water flows preferentially along channels and around the soil matrix; meanwhile, water exchanges between

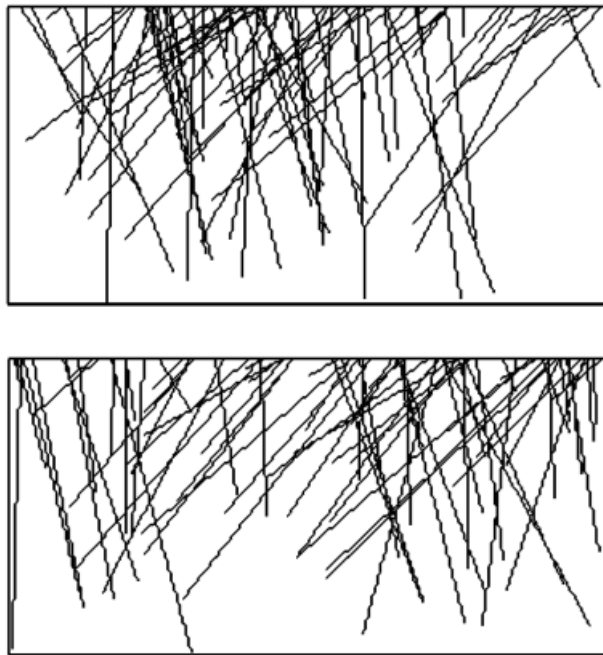


Figure 8. Two different groups of flow channels with the same parameter for a 2.0 m width



Figure 9. Distribution of channel network

the flow channel and matrix domains. Judged from the analysis above, it can be concluded that solution travels along flow channels, which may result in pollution of underground water.

Figure 11 is a sketch of a realisation of the cross-channel network calculation. This image describes the influence of the cross-channel network, which is solved by adopting the integral method. Then equations of flow movement in the soil matrix and in the irregular water network are coupled, which solves the problems of the flow mutual influence caused by the cross-channel network, and improves the computational efficiency.

5.2 Parameter sensibility analysis

In order to examine the model response to changes of specific input data, a sensibility analysis of the model was performed. In the process of establishing the network channel, the angle, length and width of the channel are the main parameters. In this paper, the parameter used in the simulation was plus or minus 10% (Table 3) and then analysis of the parameter sensibility. Based on the simulation, the sensibility analysis was performed to assess the effect produced by a given variation of the input data on the model output, with an emphasis on the influence of soil moisture. The method of sensibility analysis is to change each parameter value of the flow networks, and to observe changes in the simulation results. In a parameter sensibility analysis, other parameters remain unchanged.

As can be seen from Table 3, channel angle, length and width have little effect on the results of soil moisture, while aperture has a major influence on the simulation results. The model shows the highest sensibility to aperture. Therefore, a proper assessment of channel width is very important for the rational use of the model because these parameters can rarely be measured directly and are mostly adopted according to experience.

5.3 Effects of soil macropores on soil water flow simulation

With the same parameters in the matrix district, the comparison between Richards' equation, which assumes a continuous medium, and the hydrogeosphere model at $T = 2$ days is shown in Figure 12. It can be seen from Figure 12(a) that as infiltration water redistribution in the soil depends on the flow area and change of soil moisture in the flow area, it is clear that the proposed model produces better results on the simulation of flow patterns compared to the continuity model. As seen from the soil moisture comparison shown in Figure 12(b), with the same parameters in the matrix area, the flow network model is better than the Richards' equation, which assumes a continuous medium. The amount of water flowing into the soil and simulated by the network model is greater than that simulated by the continuum assumption of the Richards' equation model. This is mainly because the flow in soil macropores and other fast channels is taken into account in the water network model, wherein more water flows along the priority network channel, and reaches the deeper soil quickly. From water balance calculations, it can be shown that the flow network model provides a better fit to

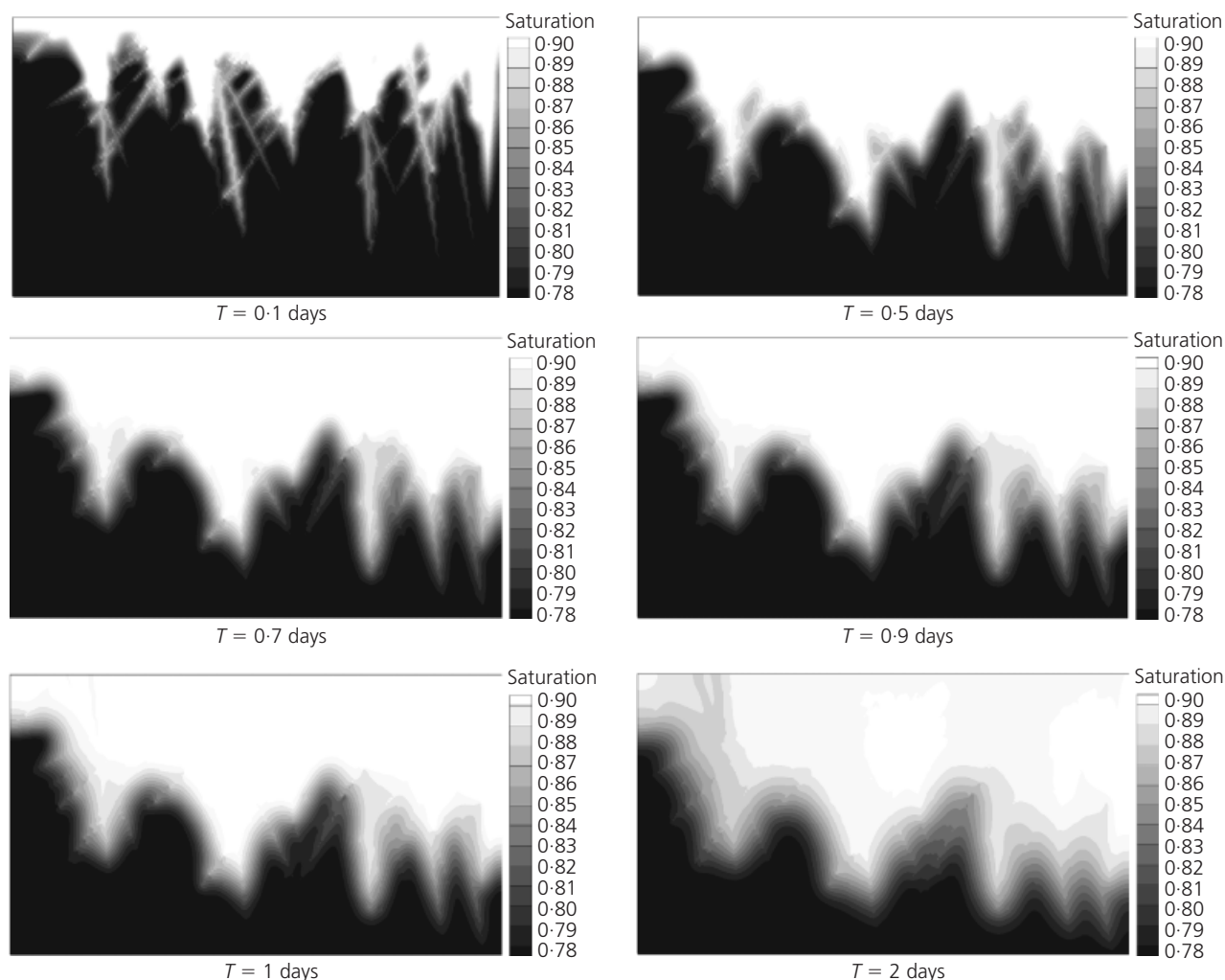


Figure 10. Dynamic process of soil water movement

experimental results, whereas the simulation results of Richards' equation model are much smaller. The flow network model is also shown to be able to simulate preferential flow more effectively, and its simulation results are better than the continuum assumption of Richards' equation.

6. Conclusions

Heterogeneous water flow in soils was investigated through dye infiltration experiments. According to the principle that iodide

ions have similar transport properties to Br^- because of their chemical similarity, and that with the unique reaction between I^- and starch, I^- ion is easy to visualise, field tracer experiments of heterogeneous flow in clay were designed and conducted under different boundaries (irrigated water depth) and different measurement scales.

On the basis of experimental image analysis, the soil medium was divided into matrix and preferential channel networks, which are formed by the network structure of flow channels in terms of space. The irregular flow network is established through random generation of angles, lengths, widths and location coordinates of flow channels, in order to achieve an accurate description of preferential flow channels. Differently structured but statistically equivalent networks have also been investigated; however, these numerous fractures have a stabilising effect on the variances of soil moisture.

Simulated results of the soil profile moisture compared well with experimental results. Images of soil profiles at different times revealed that flow patterns vary with different boundary condi-



Figure 11. The realisation of cross-channel network calculation

Parameters and results		1	2	3	4	5	6
Orientation	Middle of the smallest and largest orientation class: degrees	50–140	70–160	60–150	60–150	60–150	60–150
	μ_1, μ_2 : degrees	70, 125	90, 145	80, 135	80, 135	80, 135	80, 135
Length	λ_1	0.1	0.1	0.05	0.2	0.1	0.1
Aperture	Middle of the smallest and largest aperture class: μm	50–300	50–300	50–300	50–300	50–150	100–350
	λ_2	9000	9000	9000	9000	8100	9900
Soil moisture variance (absolute value)		2.26%	0.95%	1.78%	1.49%	4.08%	0.37%

Table 3. Parameter sensibility analysis of water flow channel

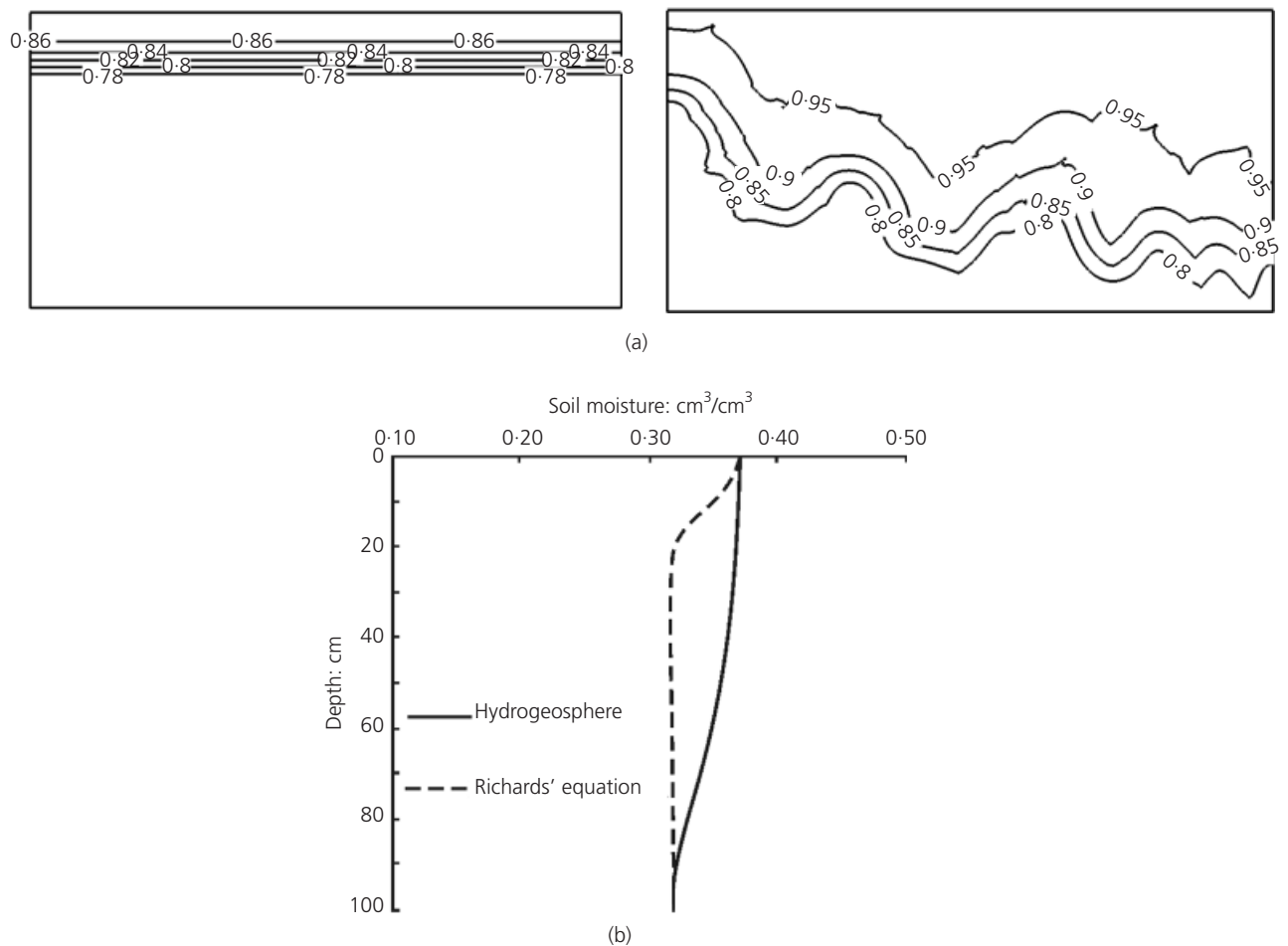


Figure 12. Comparison of simulation results: (a) comparison of water movement pattern; (b) comparison of soil moisture

tions. The flow patterns were mainly small-area preferential flow with small amounts of solution, but the flow patterns transformed from large-area preferential flow to part-matrix flow and preferential flow when the amounts of water increased. The flow patterns were part-matrix flow and overall-matrix flow when the amounts of water continued to rise.

Compared with the dual-porosity model, dual-infiltration model,

kinematic wave model and the two-phase model, the hydrogeosphere model can describe water transport at different times, the water content distribution of all domains, and the dynamic process of infiltration.

Due to the complexity of the problem, areas in need of further study include the formation of the network channel under different boundary conditions, soil texture, the importance of certain fluxes

such as evaporation during field experiments, and the effective measurement of water circulation in preferential channels.

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REFERENCES

- Aeby P, Forrer J, Steinmeier C and Fluhler H (1997) Image analysis for determination of dye tracer concentrations in sand columns. *Soil Science Society of America Journal* **61**(1): 33–35.
- Baveye P, Boast CW and Ogawa M (1998) Influence of image resolution and threshold on the apparent mass fractal characteristics of preferential flow patterns in field soils. *Water Resources Research* **34**(11): 2783–2796.
- Beven K (1991) Modeling preferential flow: an uncertain future? In *Preferential Flow* (Gish TJ and Shirmohannadi A (eds)). ASAE, MI, USA, pp. 1–11.
- Bowman RS (1984) Evaluation of some new tracers for soil water studies. *Soil Science Society of America Journal* **48**(5): 987–993.
- Forrer J, Parrita A, Kasteel R, Fluhler H and Luca D (2000) Quantifying dye tracers in soil profiles by image processing. *European Journal of Soil Science* **51**(2): 313–322.
- Gerke HH and van Genuchten MT (1993) A dual-porosity model for simulating the preferential movement of water and solutes in structured porous media. *Water Resources Research* **29**(2): 305–319.
- Glass RJ, Brainard JR and Jim Yeh TC (2005) Infiltration in unsaturated layered fluvial deposits at Rio Bravo: macroscopic anisotropy and heterogeneous transport. *Vadose Zone Journal* **4**(1): 22–31.
- Graf T and Therrien R (2007) Variable-density groundwater flow and solute transport in irregular 2D fracture networks. *Advances in Water Resources* **30**(3): 455–468.
- Green WH and Ampt GA (1911) Studies on soil physics. I. The flow of water and air through soils. *Journal of Agricultural Science* **4**: 1–24.
- Hendrickx JMH and Flury M (2001) Uniform and preferential flow, mechanisms in the vadose zone. In *Conceptual Models of Flow and Transport in the Fractured Vadose Zone*. National Research Council, National Academy Press, Washington, DC, USA, pp. 149–187.
- Jarvis N (1994) *The MACRO Model (Version 3.1). Technical Description and Sample Simulations*. Department of Soil Science, Swedish University of Agricultural Science, Uppsala, Sweden, Report and Dissertation 19.
- Kasteel R, Vogel HJ and Roth K (2002) Effect of non-linear adsorption on the transport behavior of Brilliant Blue in a field soil. *European Journal of Soil Science* **53**(2): 231–240.
- Lawes JB, Gilbert JH and Warington R (1881) On the amount and composition of the rain and drainage-waters collected at Rothamsted, Parts I and II. *Journal of the Royal Agricultural Society of England, London* **17**: 241–279.
- Morris C and Mooney SJ (2004) A high-resolution system for the quantification of preferential flow in undisturbed soil using observations of tracers. *Geoderma* **118**(1–2): 133–143.
- Nobles MM, Wilding LP and Lin HS (2010) Flow pathways of bromide and Brilliant Blue FCF tracers in caliche soils. *Journal of Hydrology* **393**(1–2): 114–122.
- Pruess K and Wang JSY (1987) Numerical modeling of isothermal and non-isothermal flow in unsaturated fractured rock – a review. In *Geophysics Monograph* (Evans DD and Nicholson TJ (eds)). American Geophysical Union, Washington, DC, USA, Geophysics Monograph vol. 42, pp. 11–22.
- Simunek J, Jarvis NJ, van Genuchten MT et al. (2003) Review and comparison of models for describing non-equilibrium and preferential flow and transport in the vadose zone. *Journal of Contaminant Hydrology* **72**(1–4): 14–35.
- Therrien R, McLaren RG, Sudicky EA and Panday SM (2004) *HYDROGEOSPHERE – A Three-Dimensional Numerical Model Describing Fully-Integrated Subsurface and Surface Flow and Solute Transport*. Université Laval, University of Waterloo, Canada.
- Wang K, Zhang R and Yasuda H (2006) Characterizing heterogeneity of soil water flow by dye infiltration experiments. *Journal of Hydrology* **328**(3–4): 559–571.
- Wang K, Zhang R and Yasuda H (2009) Characterizing heterogeneous soil water flow and solute transport using information measures. *Journal of Hydrology* **370**(1–4): 109–121.
- Wang K, Zhang R and Yasuda H (2010) Heterogeneous soil water flow and macropores described with combined tracers of dye and iodine. *Journal of Hydrology* **397**(1–2): 105–117.
- Yasuda H, Berndtsson R, Persson H, Bahri A and Takuma K (2001) Characterizing preferential transport during flood irrigation of a heavy clay soil using the dye Vitasyn Bluea. *Geoderma* **100**(1–2): 49–66.

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